quantum control by effective counterdiabatic driving

Han Wang

香港科技大学

2025-04-17

Shortcuts to adiabaticity(STA)

- Achieving acceration of adiabatic evolutions.

	adiabatic evolution(Saffman et al., 2020)	shortcut to adiabaticity(Guéry-Odelin et al., 2019)
evolution time	slow varying coefficients	shorter time with external control field
precision	Strict adiabatic conditions are difficult to satisfy	the adiabatic path can be accurately tracked with highly-accurate control
Hamiltonians	depends only on origional Hamiltonian $H_0(t)$	modified hamiltonian $H_0(t) + H_{\mathrm{CD}}(t)$

- Challange: necessity of external degrees of freedom(Additional field, auxiliary energy level, non-local coupling) which is difficult to control.

== Main Contents

- Providing systematic method to approximate counterdiabatic driving Hamiltonian with initially avaliable control Hamiltonian.(Petiziol et al., 2024)
- Method: Floquet theory, periodic driving, high-frequency modulation of operators in original adiabatic Hamiltonian.
- Applications:
  - Quantum state transfer(fmod-STIRAP)
  - Bell states in circuit QED
  - STIRAP quantum gates

counterdiabatic driving

- Considering instantaneous energy $E_{n(\lambda)}$, eigenstates $|n_\lambda\rangle(\lambda = \lambda(t))$:

$$\hat{H}_0(\lambda)|n_\lambda\rangle = E_{n(\lambda)}|n_\lambda\rangle$$

and any quantum states can be written as $|\psi(\lambda)\rangle = \sum_n a_n(\lambda)|n_\lambda\rangle$. The Schordinger equation $i\frac{\partial}{\partial t}\ |\psi(\lambda)\rangle = H(\lambda)|\psi(\lambda)\rangle$ can be written as:

$$\partial_\lambda a_{n(\lambda)} = -i\left[\dot{\lambda}^{-1}E_n(\lambda) - i\langle n_\lambda|\partial_\lambda n_\lambda\rangle\right]a_{n(\lambda)} - \sum_{m\neq n}\frac{\langle n_\lambda|\partial_\lambda \hat{H}_0(\lambda)|m_\lambda\rangle}{E_m(\lambda) - E_n(\lambda)}a_m(\lambda)$$

- During the adiabatic evolution, system will stay in one instantaneous eigenstates, but the phase will accumulate:
  - Dynamic phase: $\gamma_n^d(t) = -\frac{1}{\hbar}\int_0^t E_{n(\lambda(t'))}dt'$

counterdiabatic driving

- Geometric(Berry) phase: $\gamma_n^g(t) = -i\int_{\lambda(0)}^{\lambda(t)} \langle n_\lambda | \partial_\lambda n_\lambda \rangle d\lambda$. where $i\langle n_\lambda | \partial_\lambda n_\lambda \rangle$ is berry connection.
- For the Schordinger equation, the second term includes the nonadiabatic transitions among the instantaneous eigenstates, which should be eliminated to protect the adiabaticity.
  - Method: adding conterdiabatic driving Hamiltonian $H_{\mathrm{CD}}(\lambda)$:$\hat{H} = \hat{H}_0(\lambda) + H_{\mathrm{CD}}(\lambda)$

$$\hat{H}_{\mathrm{CD}} = i\dot{\lambda} \sum_{n,m\neq n} \frac{|n_\lambda\rangle\langle n_\lambda|\partial_\lambda \hat{H}_0(\lambda)|m_\lambda\rangle\langle m_\lambda|}{E_m(\lambda) - E_n(\lambda)}$$

Two-level system

- Driven Hamiltonian:

$$\hat{H}_0(\lambda) = \lambda\hat{\sigma}_z + \Omega\hat{\sigma}_x$$

- The calculated CD HamiltonianL

$$H_{\mathrm{CD}} = f(\lambda)\hat{\sigma}_y$$

There exists new operators which were not present in original Hamiltonian. And the direct implementation is challenging.

effective counterdiabatic driving Hamiltonian

- Question: how to approximately achieve the effective shortcut to adiabaticity using control parameters at hand? (quantum control theory(Petiziol et al., 2018))
- Method:
  - The effective CD Hamiltonian $-iH_{\mathrm{CD}}(\lambda)$ belongs to the dynamic Lie algebra of the control systems(Petiziol et al., 2018).
  - Desining suitable control function $c_{i(t)}$ generate the target evolution:

$$\hat{H}(t) = \sum_{i=1}^{N} c_{i(t)} \hat{H}_i$$

Construction of effective CD Hamiltonian

- target evolution:

$$\hat{U}(t) = \mathcal{T} \exp\left(-i \int_0^t \hat{H}(t')dt'\right)$$

- Floquet engineering(Bukov et al., 2015; Eckardt, 2017; Goldman & Dalibard, 2014): deviding time τ to n time steps during which driven hamiltonian is approximately constant: $\hat{H}_{\mathrm{CD}}^{[n]} = \hat{H}_{\mathrm{CD}}\left(\mathrm{nT} + \frac{T}{2}\right)$
- $c_i(t)$ is parametrised as time-periodic with angular frequency $\omega = 2\pi/T$ using truncated Fourier series with L harmonics of ω for nth timestep(Verdeny et al., 2014)

$$c_{i(t)} = \sum_{j=1}^{L} \left[A_{\mathrm{ij}}^{[n]} \sin(j\omega t) + B_{\mathrm{ij}}^{[n]} \cos(j\omega t)\right]$$

Construction of effective CD Hamiltonian

- Applying Floquet theory in each time step(due to the time periodicity of $c_{i(t)}$): decomposed

$$\hat{U}(t) = \hat{K}(t)\exp\left(-i\hat{H}_F t\right)$$

  - $\hat{H}_F$: Floquet Hamiltonian
  - $\hat{K}(t)$: periodic operator.

Deriving Floquet Hamiltonian

- Magnus expansion:

$$\hat{H}_F^{(0)} = \frac{1}{T}\int_0^T \hat{H}(t)\mathrm{dt}$$

$$\hat{H}_F^{(1)} = -\frac{i}{2}T\int_0^T\int_0^t \left[\hat{H}(t), \hat{H}(t')\right] dt'\mathrm{dt}$$

- Periodic Hamiltonian $\hat{H}(t)$: Fourier series expansion:

$$H(t) = \sum_{n=-\infty}^{+\infty} H_n e^{in\omega t}$$

Thus,

Deriving Floquet Hamiltonian

$$\hat{H}_{F,1} = \overline{H}_0, \tag{8a}$$

$$\hat{H}_{F,2} = \sum_{m=1}^{+\infty} \frac{1}{m}[\overline{H}_m, \overline{H}_{-m}] + \sum_{m\neq 0}[\overline{H}_0, \overline{H}_m]e^{im\omega t_0}. \tag{8b}$$

imposing the matching condition $\hat{H}_F^{[n]} = \hat{H}_{\mathrm{CD}}^{[n]}$ to solve for coefficients $A_{\mathrm{ij}}^{[n]}$ and $B_{\mathrm{ij}}^{[n]}$.

$$\hat{H}_{\mathrm{eCD}}(t) = \sqrt{\frac{\omega}{2}\frac{\dot{\lambda}(t)\Omega}{\lambda(t)^2+\Omega^2}}\,[\sin(\omega t)\hat{\sigma}_z - \cos(\omega t)\hat{\sigma}_x]. \tag{6}$$

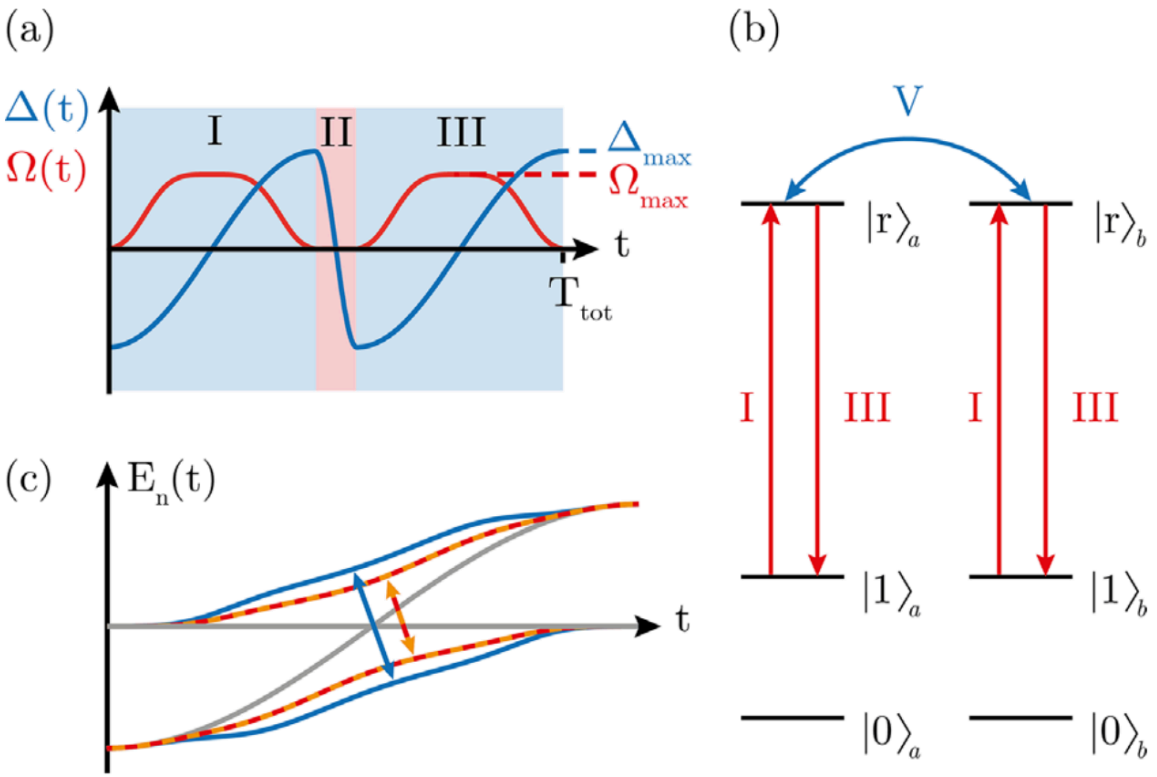
Application:frequency-modulated-STIRAP

- Due to selective rules(or some other reasons), the system cannot be directly transfered from state $|0\rangle$ to $|2\rangle$. The system can be transfered from $|0\rangle$ to $|2\rangle$ by using the intermediate state $|1\rangle$ as a bridge.
- The exact CD fields is applied to realize the transition between $|0\rangle$ and $|2\rangle$.

$$\hat{H}(t) = \frac{1}{2}\begin{bmatrix} 0 & \Omega_-(t) & i\Omega_{\mathrm{CD}}(t) \\ \Omega_-^*(t) & 2\Delta_1 & \Omega_+(t) \\ -i\Omega_{\mathrm{CD}}(t) & \Omega_+^*(t) & 0 \end{bmatrix}$$

- Effective CD Hamiltonian can be achieved through adding new Fourier components to the original Hamiltonian.(Pump and Stokes Rabi frequency).

Application: acceration of CZ gate



Reference

Bibliography

[1] M. Saffman, I. I. Beterov, A. Dalal, E. J. Páez, and B. C. Sanders, "Symmetric Rydberg controlled-Z gates with adiabatic pulses", *Phys. Rev. A*, vol. 101, no. 6, p. 62309, Jun. 2020, doi: 10.1103/PhysRevA.101.062309.

[2] D. Guéry-Odelin, A. Ruschhaupt, A. Kiely, E. Torrontegui, S. Martínez-Garaot, and J. G. Muga, "Shortcuts to adiabaticity: Concepts, methods, and applications," *Rev. Mod. Phys.*, vol. 91, no. 4, p. 45001, Oct. 2019, doi: 10.1103/RevModPhys.91.045001.

[3] F. Petiziol, F. Mintert, and S. Wimberger, "Quantum control by effective counterdiabatic driving," *Europhysics Letters*, vol. 145, 2024, [Online]. Available: https://api.semanticscholar.org/CorpusID:266855431

Reference

[4] F. Petiziol, B. Dive, F. Mintert, and S. Wimberger, "Fast adiabatic evolution by oscillating initial Hamiltonians," *Phys. Rev. A*, vol. 98, no. 4, p. 43436, Oct. 2018, doi: 10.1103/PhysRevA.98.043436.

[7] N. Goldman and J. Dalibard, "Periodically Driven Quantum Systems: Effective Hamiltonians and Engineered Gauge Fields," *Phys. Rev. X*, vol. 4, no. 3, p. 31027, Aug. 2014, doi: 10.1103/PhysRevX.4.031027.

[5] M. Bukov, L. D'Alessio, and A. P. and, "Universal high-frequency behavior of periodically driven systems: from dynamical stabilization to Floquet engineering," *Advances in Physics*, vol. 64, no. 2, pp. 139–226, 2015, doi: 10.1080/00018732.2015.1055918.

[6] A. Eckardt, "Colloquium: Atomic quantum gases in periodically driven optical lattices," *Rev. Mod. Phys.*, vol. 89, no. 1, p. 11004, Mar. 2017, doi: 10.1103/RevModPhys.89.011004.

Reference

[8] A. Verdeny, Ł. Rudnicki, C. A. Müller, and F. Mintert, "Optimal Control of Effective Hamiltonians," *Phys. Rev. Lett.*, vol. 113, no. 1, p. 10501, Jul. 2014, doi: 10.1103/PhysRevLett.113.010501.